

PVsyst - Simulation report

Grid-Connected System

Project: NOVO-CAMPINAS

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 1200 Wp

Campinas - Brazil



PVsyst V8.1.5

VCO, Simulation date:
16/09/26 11:54
with V8.1.5

Project summary

Geographical Site

Campinas

Brazil

Situation

Latitude -22.9056 °(S)
Longitude -47.0596 °(W)
Altitude 685 m
Time zone UTC-3

Project settings

Albedo 0.20

Weather data

Campinas

NASA POWER satellite data 2001-2020 - Synthetic

System summary

Grid-Connected System

No 3D scene defined, no shadings

Orientation #1

Horizontal plane

Near Shadings

no Shadings

User's needs

Unlimited load (grid)

System information

PV Array

Nb. of modules

2 units

Pnom total

1200 Wp

Inverters

Nb. of units

Total power

Pnom ratio

1 unit

1.5 kWac

0.80

Results summary

Simulation step Hourly

Produced Energy 1945.4 kWh/year Specific production 1621 kWh/kWp/year Perf. Ratio PR 86.41 %

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General parameters

Grid-Connected System

No 3D scene defined, no shadings

Simulation step

Hourly

Orientation #1

Horizontal plane

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

no Shadings

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer Generic
Model JKM-600N-72HL4-V
(Original PVsyst database)
Unit Nom. Power 600 Wp
Number of PV modules 2 units
Nominal (STC) 1200 Wp
Modules 1 unit x 2 In series

At operating cond. (50°C)

Pmpp 1113 Wp
U mpp 82 V
I mpp 14 A

Total PV power

Nominal (STC) 1.20 kWp
Total 2 modules
Module area 5.2 m²
Cell area 4.8 m²

Inverter

Manufacturer Generic
Model MIC 1500TL-X
(Original PVsyst database)
Unit Nom. Power 1.50 kWac
Number of inverters 1 unit
Total power 1.5 kWac
Operating voltage 50-500 V
Pnom ratio (DC:AC) 0.80

Total inverter power

Total power 1.5 kWac
Number of inverters 1 unit
Pnom ratio 0.80

Array losses

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 98 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss

Loss Fraction -0.75 %

Module mismatch losses

Loss Fraction 0.50 % at MPP

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000



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Main results

Simulation step

Hourly

System Production

Produced Energy

1945.4 kWh/year

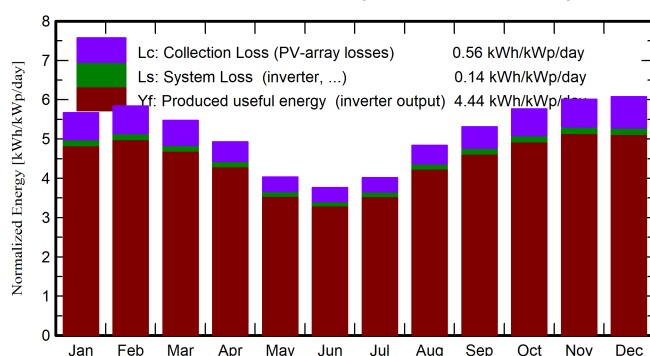
Specific production

1621 kWh/kWp/year

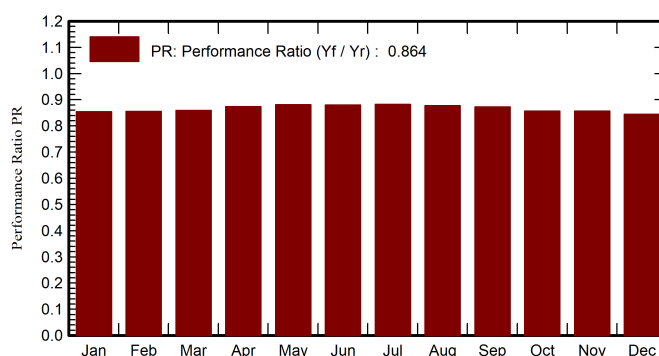
Perf. Ratio PR

86.41 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	kWh	kWh	ratio
January	175.8	83.60	22.84	175.7	171.5	185.7	180.0	0.854
February	163.6	66.87	22.90	163.5	159.5	173.0	167.8	0.855
March	169.7	64.91	22.14	169.7	165.1	180.2	174.8	0.859
April	147.9	47.79	20.65	147.9	143.3	159.8	155.0	0.873
May	125.2	41.71	17.76	125.1	120.6	136.5	132.2	0.880
June	112.9	33.95	17.03	112.8	108.0	123.0	119.1	0.880
July	124.7	34.13	16.80	124.6	119.5	136.2	132.0	0.883
August	150.0	36.87	18.37	150.0	145.2	162.7	157.9	0.877
September	159.3	49.12	20.66	159.2	155.2	171.8	166.7	0.872
October	178.9	69.89	22.20	178.8	174.4	189.3	183.7	0.856
November	180.4	78.40	22.07	180.4	176.0	191.3	185.5	0.857
December	188.3	85.91	22.83	188.3	183.8	196.7	190.7	0.844
Year	1876.7	693.16	20.51	1876.0	1822.1	2006.1	1945.4	0.864

Legends

GlobHor Global horizontal irradiation

DiffHor Horizontal diffuse irradiation

T_Amb Ambient Temperature

GlobInc Global incident in coll. plane

GlobEff Effective Global, corr. for IAM and shadings

EArray Effective energy at the output of the array

E_Grid Energy injected into grid

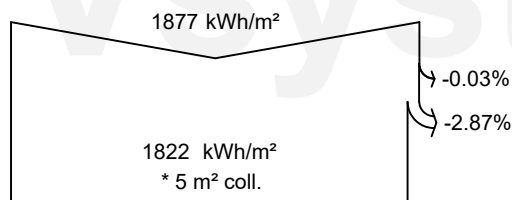
PR Performance Ratio



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Loss diagram



Global horizontal irradiation

Global incident in coll. plane

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

Module array mismatch loss

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

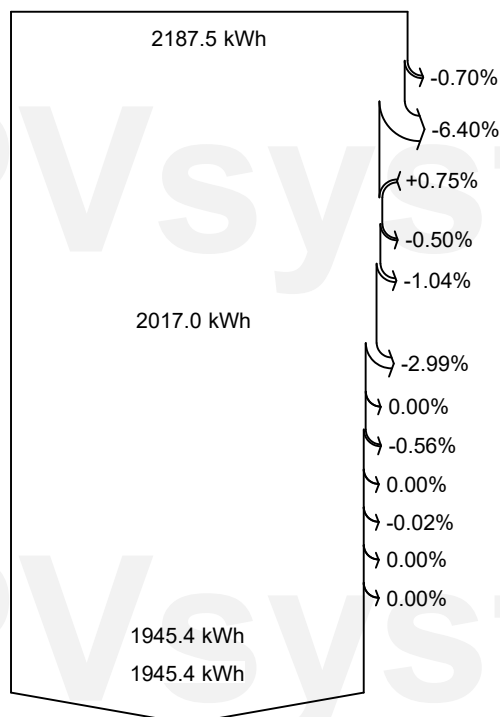
Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

Energy injected into grid

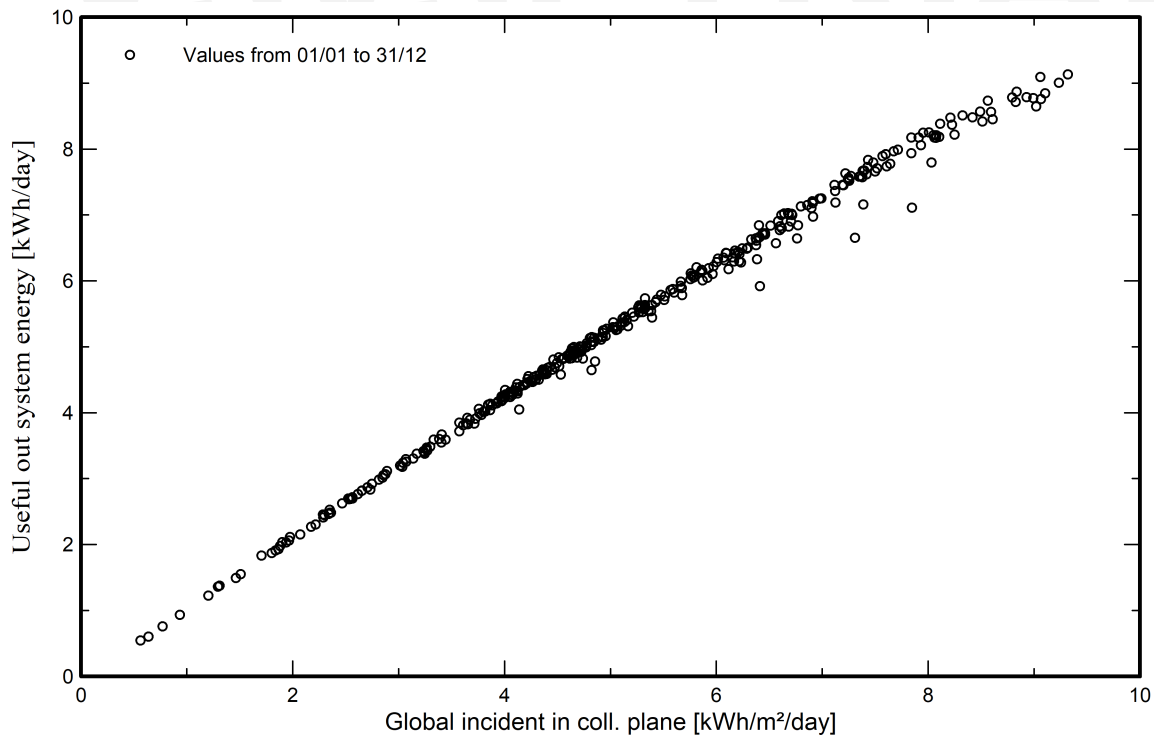




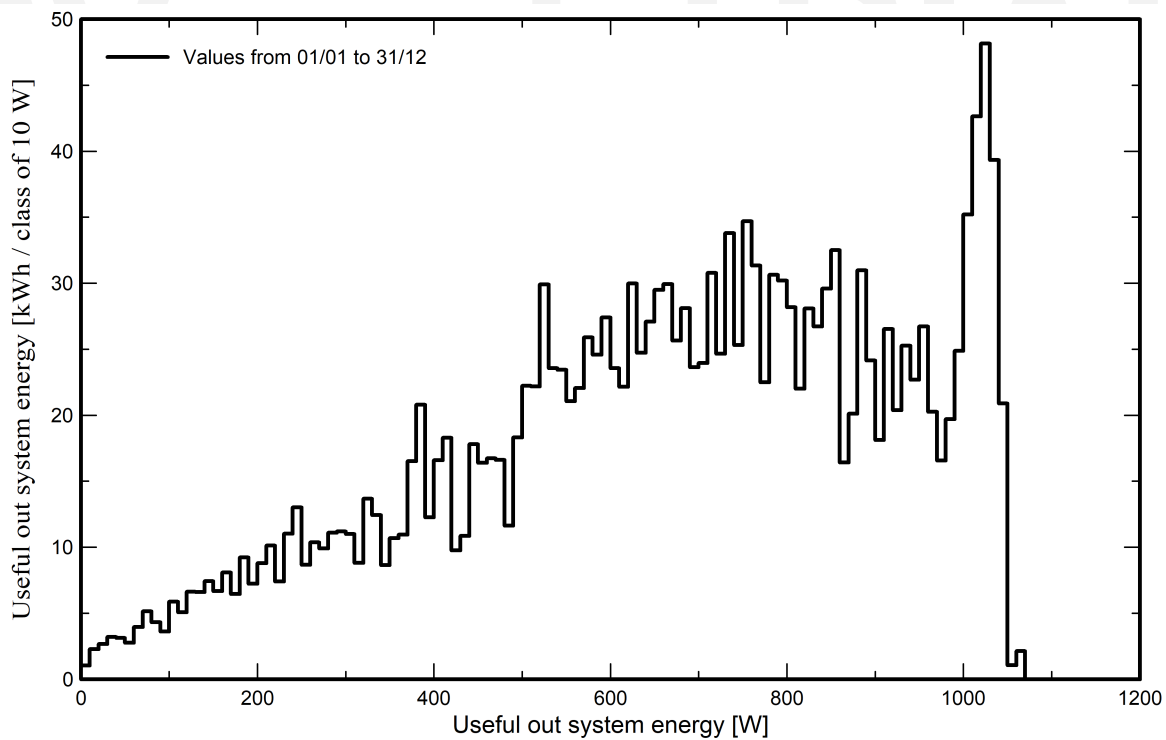
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Predef. graphs
Daily Input/Output diagram



System Output Power Distribution

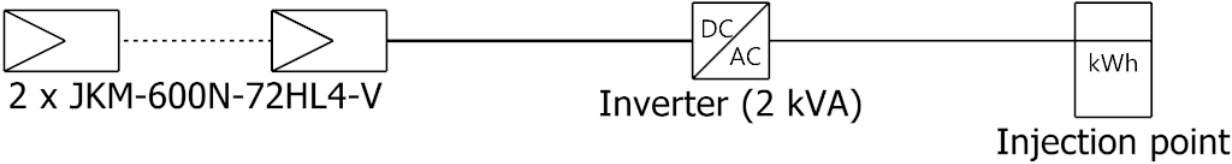




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Single-line diagram



PV module	JKM-600N-72HL4-V
Inverter	MIC 1500TL-X
String	2 x JKM-600N-72HL4-V

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